

1.-

- (a) The area of each face is $A = 1.00 \text{ m}^2$.

For the left face, the angle between the electric field and the normal is 0° :

$$\begin{aligned}(\Phi_E)_{\text{left face}} &= EA \cos \theta = (20.0 \text{ N/C})(1.00 \text{ m}^2) \cos 0^\circ \\ &= 20.0 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

For the right face, the angle between the electric field and the normal is 180° :

$$\begin{aligned}(\Phi_E)_{\text{right face}} &= EA \cos \theta = (35.0 \text{ N/C})(1.00 \text{ m}^2) \cos 180^\circ \\ &= -35.0 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

For the top face, the angle between the electric field and the normal is 180° :

$$\begin{aligned}(\Phi_E)_{\text{top face}} &= EA \cos \theta = (25.0 \text{ N/C})(1.00 \text{ m}^2) \cos 180^\circ \\ &= -25.0 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

For the bottom face, the angle between the electric field and the normal is 0° :

$$\begin{aligned}(\Phi_E)_{\text{bottom face}} &= EA \cos \theta = (15.0 \text{ N/C})(1.00 \text{ m}^2) \cos 0^\circ \\ &= 15.0 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

For the front face, the angle between the electric field and the normal is 0° :

$$\begin{aligned}(\Phi_E)_{\text{front face}} &= EA \cos \theta = (20.0 \text{ N/C})(1.00 \text{ m}^2) \cos 0^\circ \\ &= 20.0 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

For the back face, the angle between the electric field and the normal is 0° :

$$\begin{aligned}(\Phi_E)_{\text{back face}} &= EA \cos \theta = (20.0 \text{ N/C})(1.00 \text{ m}^2) \cos 0^\circ \\ &= 20.0 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

The total flux is then

$$\begin{aligned}\Phi_E &= (20.0 - 35.0 - 25.0 + 15.0 + 20.0 + 20.0) \text{ N} \cdot \text{m}^2/\text{C} \\ &= \boxed{15.0 \text{ N} \cdot \text{m}^2/\text{C}}\end{aligned}$$

- (b) $\Phi_E = \frac{q_{\text{in}}}{\epsilon_0} \rightarrow q_{\text{in}} = \epsilon_0 \Phi_E = (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(15.0 \text{ N} \cdot \text{m}^2/\text{C})$
 $= \boxed{1.33 \times 10^{-10} \text{ C}}$

2.-

- (a) Take the origin at the point where we will find the potential. One ring, of width dx , has charge $\frac{Qdx}{h}$ and, according to Example 25.5, creates potential

$$dV = \frac{k_e Q dx}{h \sqrt{x^2 + R^2}}$$

The whole stack of rings creates potential

$$\begin{aligned} V &= \int_{\text{all charge}} dV = \int_d^{d+h} \frac{k_e Q dx}{h \sqrt{x^2 + R^2}} = \frac{k_e Q}{h} \ln \left(x + \sqrt{x^2 + R^2} \right) \Big|_d^{d+h} \\ &= \boxed{\frac{k_e Q}{h} \ln \left(\frac{d+h + \sqrt{(d+h)^2 + R^2}}{d + \sqrt{d^2 + R^2}} \right)} \end{aligned}$$

- (b) A disk of thickness dx has charge $\frac{Qdx}{h}$ and charge-per-area $\frac{Qdx}{\pi R^2 h}$. According to Example 25.6, it creates potential

$$dV = 2\pi k_e \frac{Q dx}{\pi R^2 h} \left(\sqrt{x^2 + R^2} - x \right)$$

Integrating,

$$\begin{aligned} V &= \int_d^{d+h} \frac{2k_e Q}{R^2 h} \left(\sqrt{x^2 + R^2} dx - x dx \right) \\ &= \frac{2k_e Q}{R^2 h} \left[\frac{1}{2} x \sqrt{x^2 + R^2} + \frac{R^2}{2} \ln \left(x + \sqrt{x^2 + R^2} \right) - \frac{x^2}{2} \right]_d^{d+h} \\ V &= \boxed{\frac{k_e Q}{R^2 h} \left[(d+h) \sqrt{(d+h)^2 + R^2} - d \sqrt{d^2 + R^2} \right.} \\ &\quad \left. - 2dh - h^2 + R^2 \ln \left(\frac{d+h + \sqrt{(d+h)^2 + R^2}}{d + \sqrt{d^2 + R^2}} \right) \right]} \end{aligned}$$

- (a) In an empty universe, the 20.0-nC charge can be placed at its location with no energy investment. At a distance of 4.00 cm, it creates a potential

$$V_1 = \frac{k_e q_1}{r} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(20.0 \times 10^{-9} \text{ C})}{0.0400 \text{ m}} = 4.50 \text{ kV}$$

To place the 10.0-nC charge there we must put in energy

$$U_{12} = q_2 V_1 = (10.0 \times 10^{-9} \text{ C})(4.50 \times 10^3 \text{ V}) = 4.50 \times 10^{-5} \text{ J}$$

Next, to bring up the -20.0-nC charge requires energy

$$\begin{aligned} U_{23} + U_{13} &= q_3 V_2 + q_3 V_1 = q_3 (V_2 + V_1) \\ &= (-20.0 \times 10^{-9} \text{ C})(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \\ &\quad \times \left(\frac{10.0 \times 10^{-9} \text{ C}}{0.0400 \text{ m}} + \frac{20.0 \times 10^{-9} \text{ C}}{0.0800 \text{ m}} \right) \\ &= -4.50 \times 10^{-5} \text{ J} - 4.50 \times 10^{-5} \text{ J} \end{aligned}$$

The total energy of the three charges is

$$U_{12} + U_{23} + U_{13} = \boxed{-4.50 \times 10^{-5} \text{ J}}$$

- (b) The three fixed charges create this potential at the location where the fourth is released:

$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ &= (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \\ &\quad \times \left(\frac{20.0 \times 10^{-9} \text{ C}}{\sqrt{(0.0400 \text{ m})^2 + (0.0300 \text{ m})^2}} \right. \\ &\quad \left. + \frac{10.0 \times 10^{-9} \text{ C}}{0.0300 \text{ m}} - \frac{20.0 \times 10^{-9} \text{ C}}{\sqrt{(0.0400 \text{ m})^2 + (0.0300 \text{ m})^2}} \right) \\ V &= 3.00 \times 10^3 \text{ V} \end{aligned}$$

Energy of the system of four charged objects is conserved as the fourth charge flies away:

$$\begin{aligned} \left(\frac{1}{2} m v^2 + qV \right)_i &= \left(\frac{1}{2} m v^2 + qV \right)_f \\ 0 + (40.0 \times 10^{-9} \text{ C})(3.00 \times 10^3 \text{ V}) &= \frac{1}{2} (2.00 \times 10^{-13} \text{ kg}) v^2 + 0 \\ v &= \sqrt{\frac{2(1.20 \times 10^{-4} \text{ J})}{2 \times 10^{-13} \text{ kg}}} = \boxed{3.46 \times 10^4 \text{ m/s}} \end{aligned}$$

Solution: Using Gauss's law, the electrostatic field has the following values:

$$\vec{E}(\vec{r}) = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{q}{r^2} \hat{r}$$

$$r < r_1$$

and

$$r > r_2$$

$$E(r) = 0$$

$$r_1 \leq r \leq r_2 .$$

The electrostatic energy of this system is calculated using

$$U = \int_V \frac{1}{2} \epsilon_0 E^2 dV$$

and subtracting out the self-energy of the charge q . For a sphere,

$$dv = 4\pi r^2 dr$$

therefore

$$U = (4\pi\epsilon_0) \frac{1}{8\pi} \left\{ \int_0^{r_1} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right)^2 4\pi r^2 dr + \int_{r_2}^{\infty} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right)^2 4\pi r^2 dr - \int_0^{\infty} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \right)^2 4\pi r^2 dr \right\} .$$

$$U = \frac{1}{4\pi\epsilon_0} \frac{1}{2} q^2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right) .$$